MolPallete — Corpus EDA

naveja_recap

/home/mogan/preprocessed/molpallete/coconut-flavordb_v4/naveja_recap

generated 2026-08-20 12:36

A · Provenance

field	value
sources	['flavordb', 'coconut']
method	naveja_recap
decomposition_params	{'ratio': 0.3333333333333333, 'include_ring': True, 'max_cores':
core_ratio	0.3333333333333333
max_cores	4
max_rgroups	8
keep_stereo	True
neutralise	False
dedup	True
n_duplicates_removed	4145
size_filter	{'min_heavy_atoms': 5, 'max_heavy_atoms': 50}
graph_hash_version	3
rdkit_version	2026.03.2
created_at	2026-08-20T12:34:42

B · Composition

quantity	value
molecules	393,066
from coconut	372,254 (94.7%)
from flavordb	20,812 (5.3%)
decompositions	1,145,758
per molecule	mean 2.91, median 3, max 4
R-group slots	1,608,324
per decomposition	mean 1.40
k >= 2	31.03% of decompositions

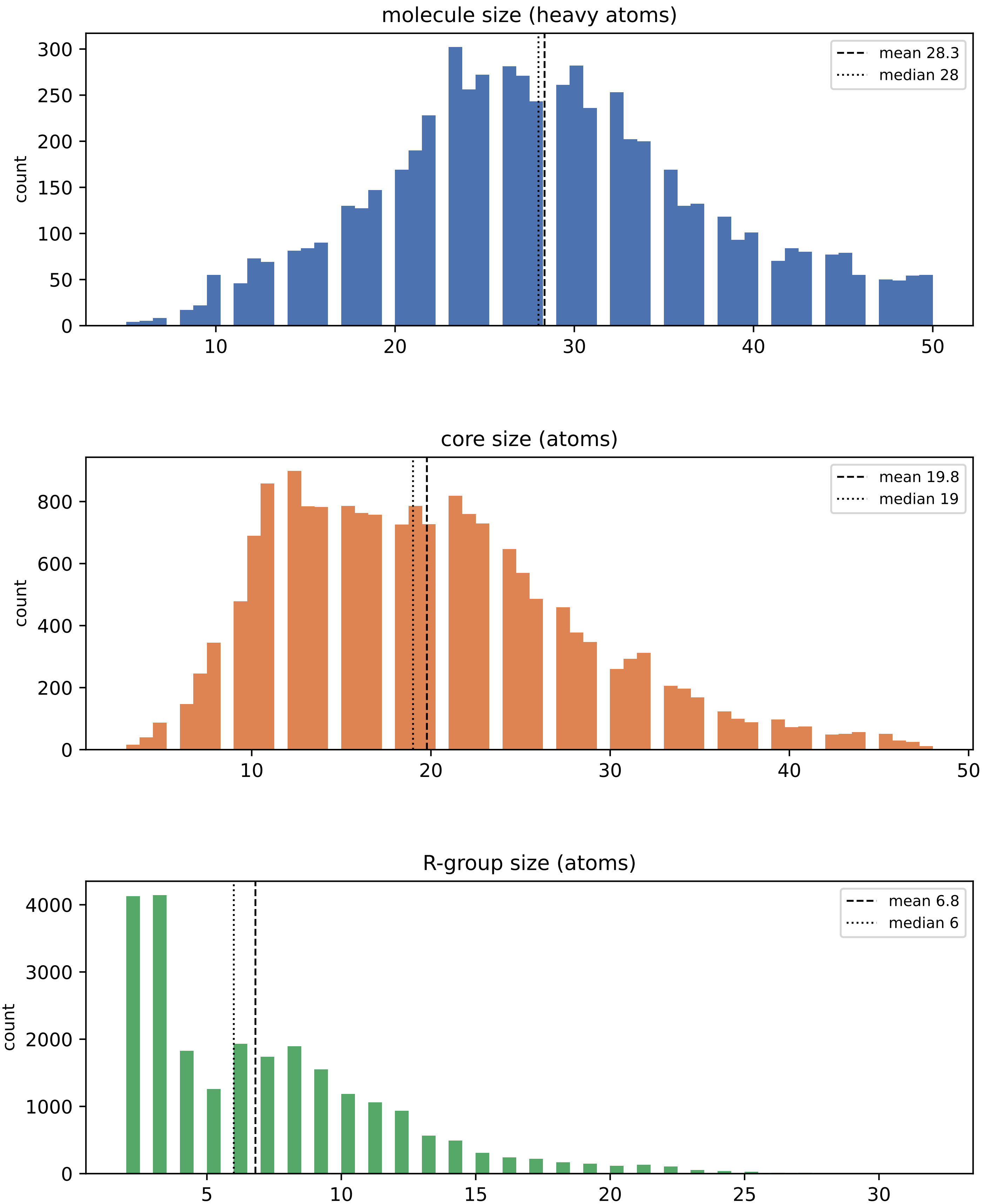
k>=2 is what the islinked subset space and the core-decoration objective have to work with.

C · Augmentation space

scheme	size
sampling_unit = molecule	393,066 instances/epoch
sampling_unit = decomposition	1,145,758 instances/epoch (2.9x)
full (mol, core, islinked) space	2,508,474 (6.4 per molecule)

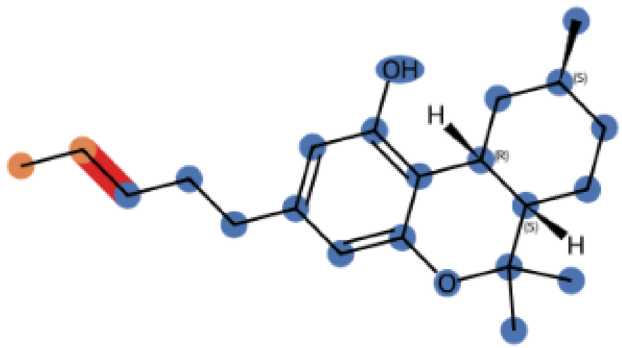
MolPLA's two stages: one molecule yields many cores; one core yields 2^k - 1 islinked subsets.

D · Size distributions

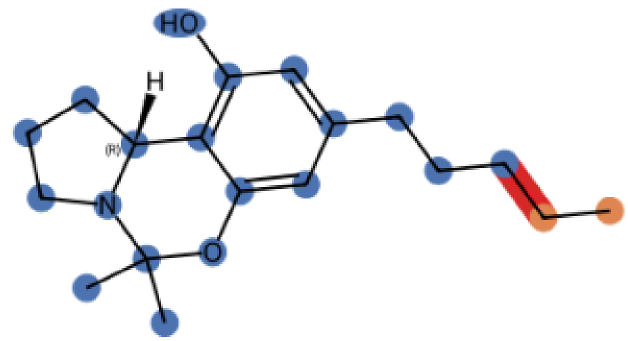


E · Example decompositions (ratio 0.3333333333333333)

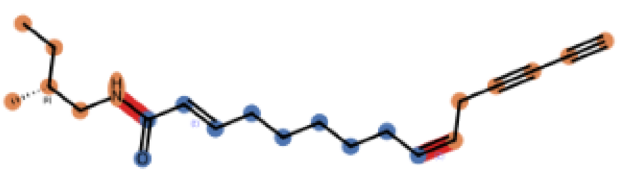
CNP0000075   k=1   core 21 atoms, R-groups [2]



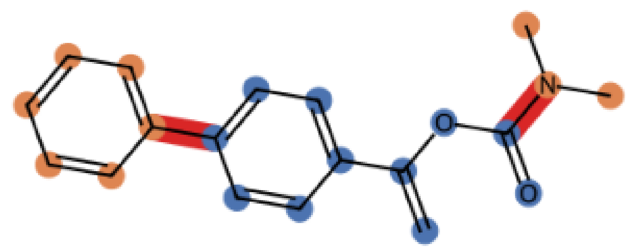
CNP0000164   k=1   core 19 atoms, R-groups [2]



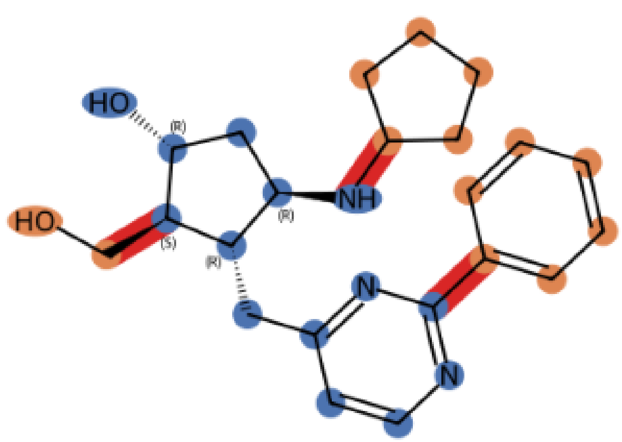
CNP0000306   k=2   core 10 atoms, R-groups [6, 6]



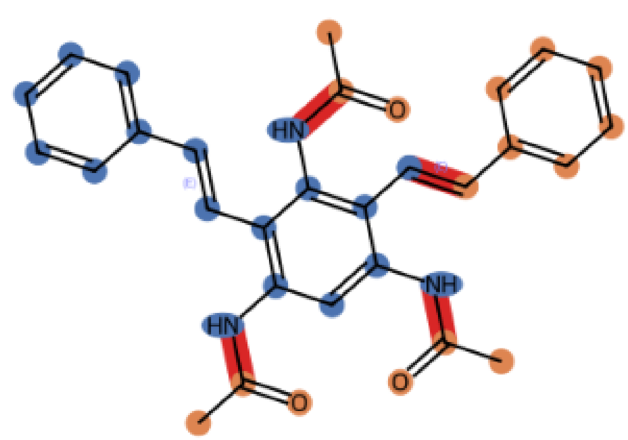
CNP0000665   k=2   core 11 atoms, R-groups [3, 6]



CNP0005977   k=3   core 14 atoms, R-groups [2, 6, 5]

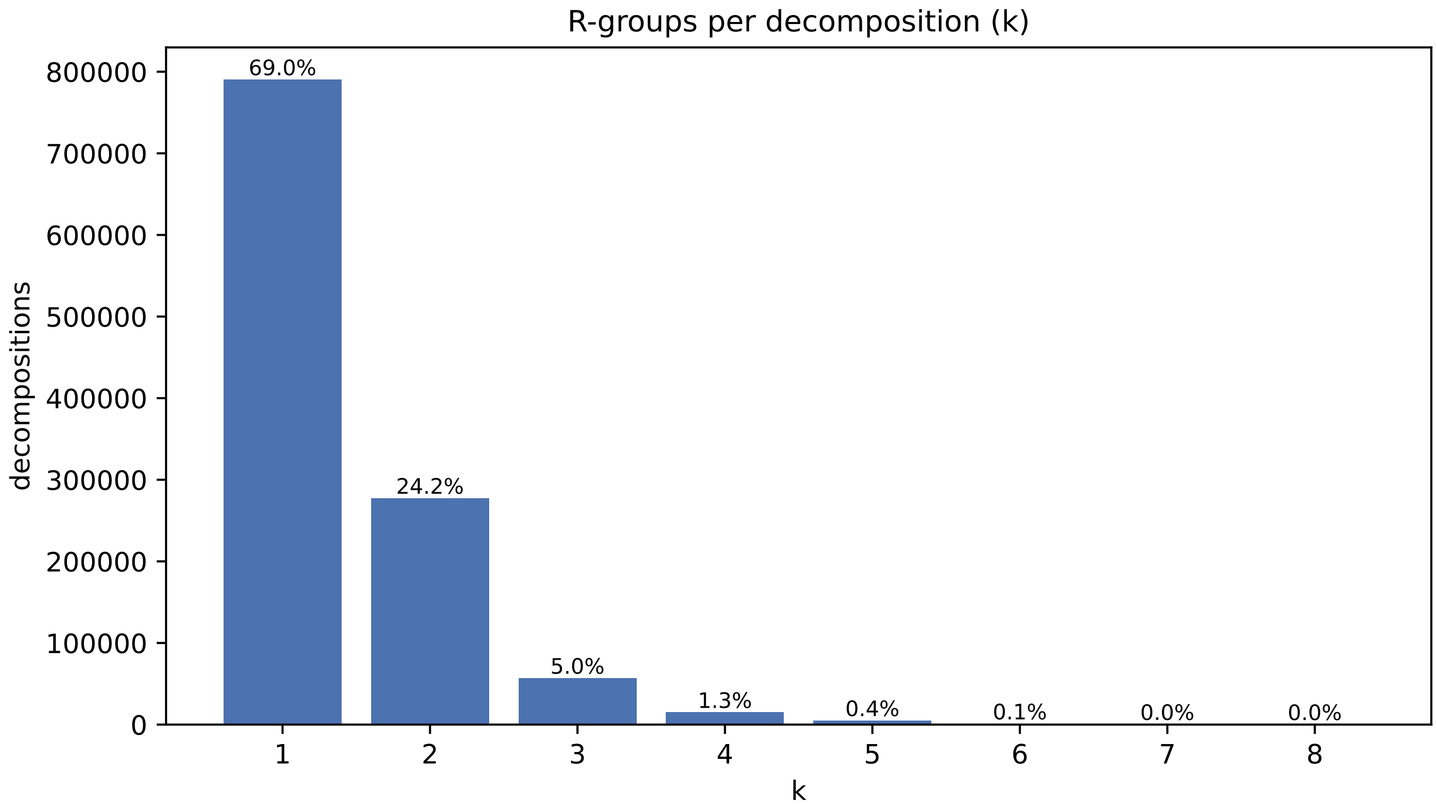


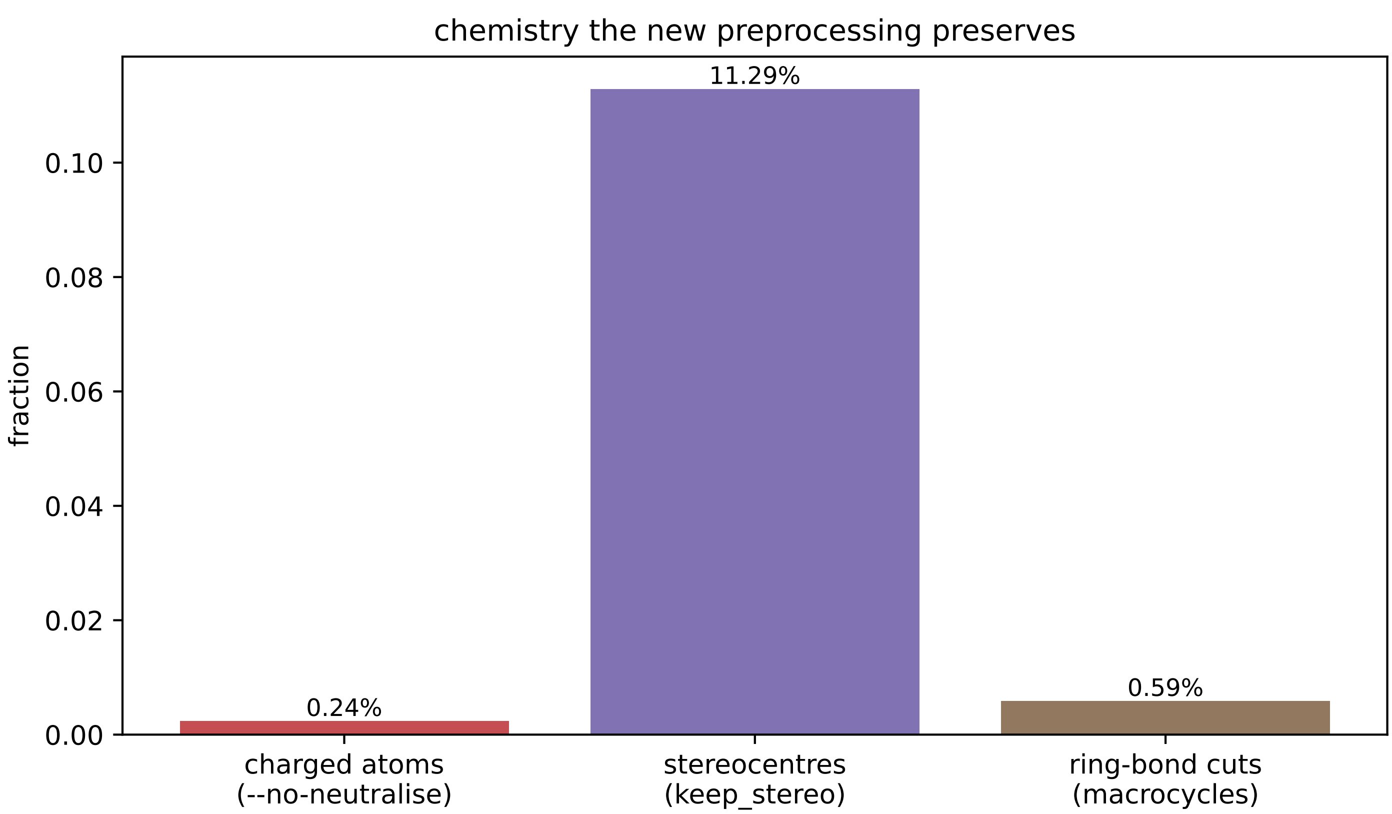
CNP0070646   k=4   core 18 atoms, R-groups [3, 3, 3, 7]



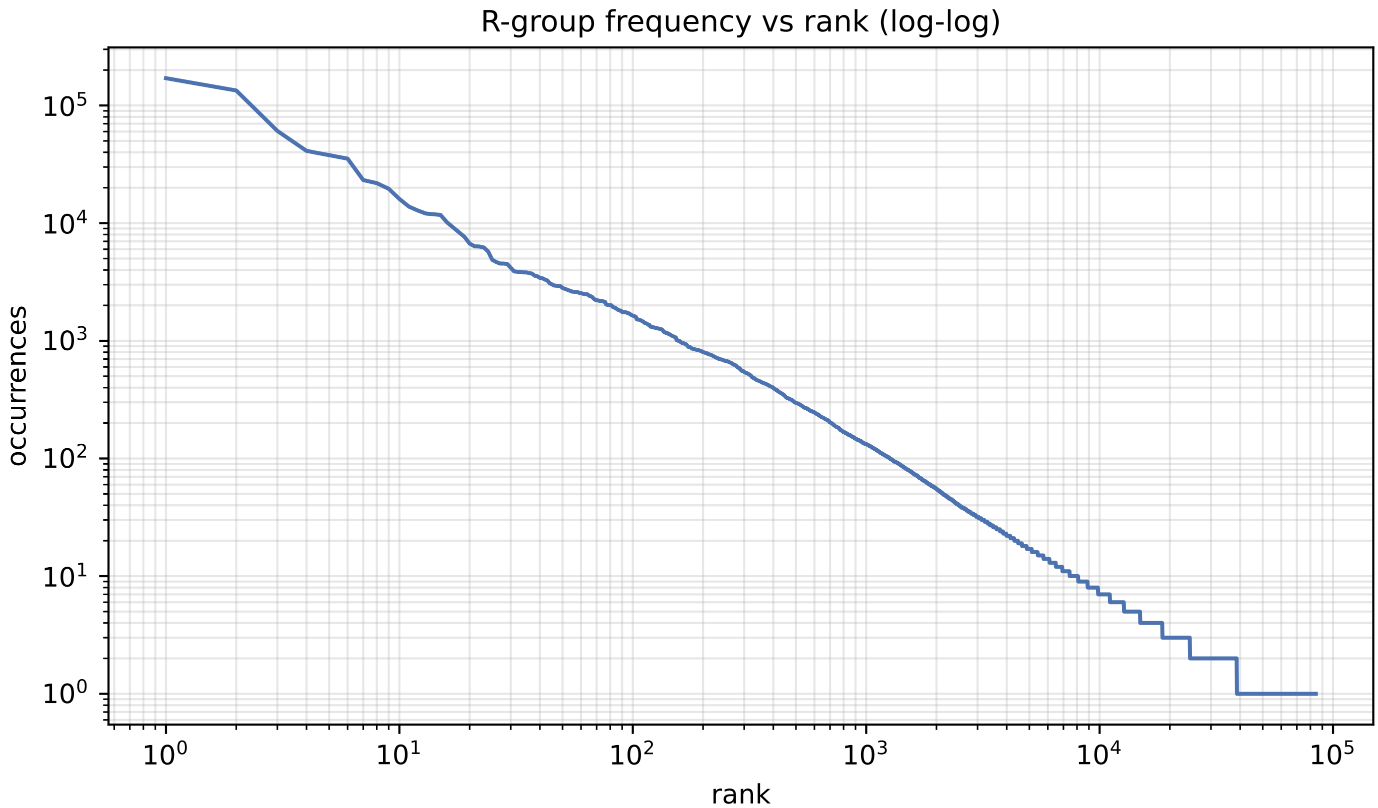
blue = core   orange = R-groups   red = cut bonds

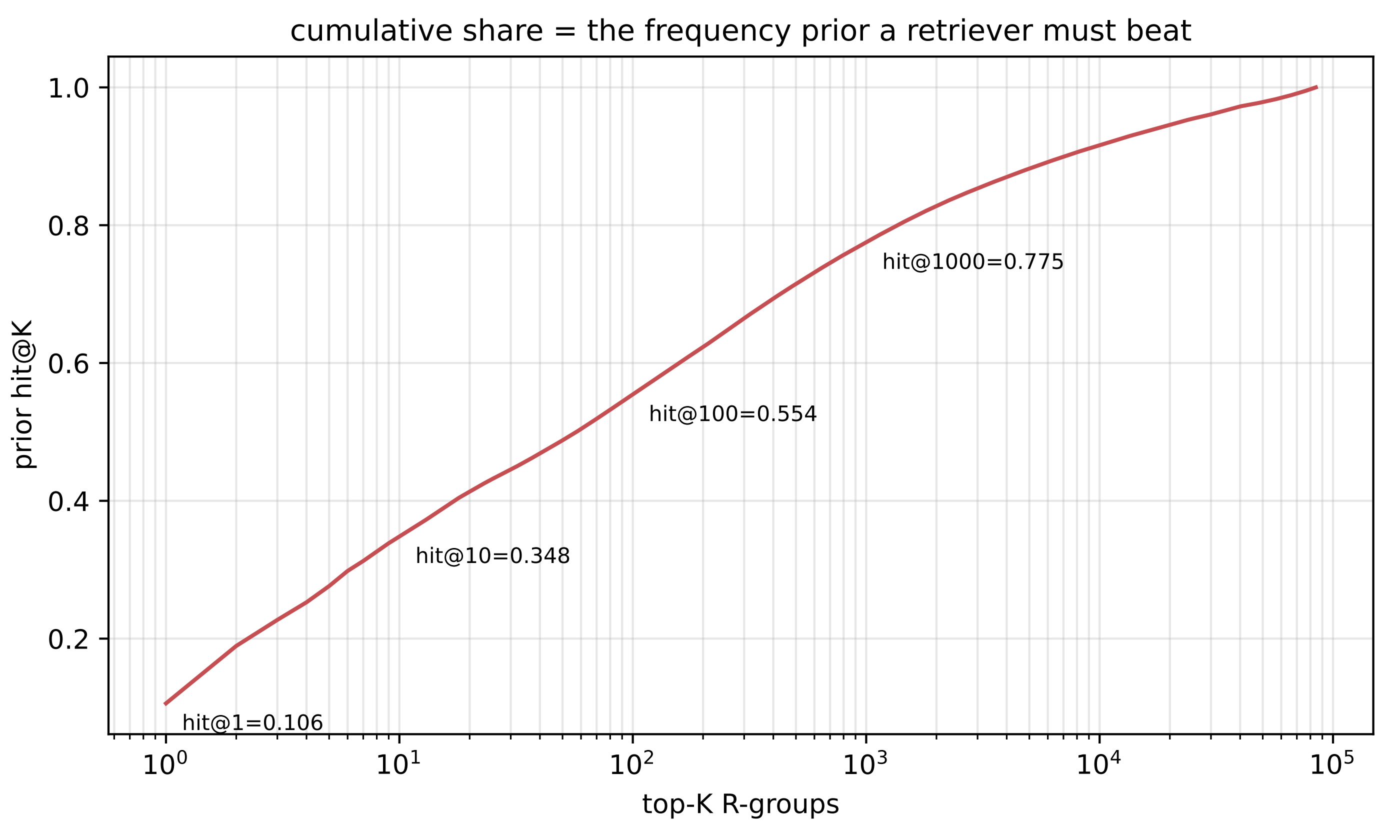
F · Decomposition shape & preserved chemistry





G · R-group library skew





H · R-group library

quantity	value
distinct R-groups	84,387
occurrences	1,608,324
effective size (exp H)	800
top-1 share	10.60%
top-10 cover	34.8%
top-100 cover	55.4%

Effective size, not the raw count, is the number of classes retrieval actually chooses between.

I · Most frequent R-groups

count	share	SMILES
170,556	10.60%	*OC
134,073	8.34%	*C(C)=O
60,791	3.78%	*C(C)C
41,125	2.56%	*CC
37,816	2.35%	*c1ccccc1
35,317	2.20%	*CO
23,237	1.44%	*OC(C)=O
21,872	1.36%	*C(=O)O
19,595	1.22%	*Cc1ccccc1
16,052	1.00%	*CCC
13,799	0.86%	*O[C@@H]1O[C@H](CO)[C@@H](O)[C@H](O)[C
12,792	0.80%	*C(=O)c1ccccc1
12,062	0.75%	*c1ccc(OC)cc1
11,866	0.74%	*OCC
11,749	0.73%	*CC(=O)O
10,153	0.63%	*c1ccc(O)cc1
9,171	0.57%	*CCO
8,387	0.52%	*C=O
7,647	0.48%	*CCCC
6,703	0.42%	*C1O[C@H](CO)[C@@H](O)[C@H](O)[C@H]1O

J · Instance-level common-R-group filter

quantity	value
percentile	99.9
count threshold	1,862
hashes flagged common	85 of 84,387
their share of occurrences	53.8%
single-R-group instances rejected	12,886/24,292 (53.0%)
mean R-group size, rejected	4.13 atoms
mean R-group size, kept	9.83 atoms

Filters INSTANCES, not the library: the retrieval target space and its frequency prior are unchanged. At k=1 the rule reduces to 'drop if the R-group is common', so rejection is far heavier than MolPLA saw.